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Contents

CHAPTER 1 ■ FusionDxNet: Lung & Colon Cancer Classification	1
PROTIK CHAKROBORTY, PALLAB CHOWDHURY, PRITOM CHAKROBORTY, ARUN KUMAR SIKDER, and MD MOHAIMINUL ISLAM SAJIB	
1.1 INTRODUCTION	2
1.2 RELATED WORK	4
1.3 MATERIALS AND METHODS	4
1.3.1 Dataset	4
1.3.2 Method Overview	6
1.3.3 Proposed Methodology	7
1.3.4 Feature Extractor	9
1.3.5 Classification	10
1.3.5.1 Support Vector Machine:	10
1.3.5.2 Custom CNN Model	10
1.3.5.3 Voting Classifier:	10
1.4 EXPERIMENTATION AND PERFORMANCE ANALYSIS	11
1.4.1 Model's Performance Evaluation	11
1.4.1.1 Accuracy	11
1.4.1.2 F1 Score	11
1.4.1.3 Precision	11
1.4.1.4 Recall	11
1.4.2 Performance Evaluation	12
1.5 LIMITATION & FUTURE WORK	12
1.6 CONCLUSION AND FUTURE WORK	13



FusionDxNet: A Fusion-Based Approach for Robust Lung & Colon Cancer Classification Using Histopathological Images

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CONTENTS

1.1	Introduction	2
1.2	Related Work	4
1.3	Materials and methods	4
1.3.1	Dataset	4
1.3.2	Method Overview	6
1.3.3	Proposed Methodology	7
1.3.4	Feature Extractor	9
1.3.5	Classification	10
1.3.5.1	Support Vector Machine:	10
1.3.5.2	Custom CNN Model	10
1.3.5.3	Voting Classifier:	10
1.4	Experimentation and Performance Analysis	11

1.4.1	Model's Performance Evaluation	11
1.4.1.1	Accuracy	11
1.4.1.2	F1 Score	11
1.4.1.3	Precision	11
1.4.1.4	Recall	11
1.4.2	Performance Evaluation	11
1.5	Limitation & Future Work	12
1.6	Conclusion and Future Work	13

Cancer is still one among the most significant global health challenges, with lung and colorectal cancers ranking among the most commonly diagnosed types. A prompt and precise diagnosis is essential for achieving effective treatment outcomes, yet traditional diagnostic techniques are often time-consuming and reliant on expert interpretation. In this study, we introduce FusionDxNet, an advanced deep learning-based framework designed for the automated classification of lung and colon cancers from histopathological images. The proposed model leverages the strengths of two powerful pretrained convolutional neural networks—ResNet50 and MobileNet—for robust feature extraction. The extracted features are fused and passed through a hybrid architecture comprising Logistic Regression, a custom 1D CNN, and an ensemble classifier to deliver the final prediction. This multi-stage fusion approach effectively captures complex patterns across different image types, enhancing model generalizability and classification performance. Extensive experiments conducted on publicly available datasets demonstrate that FusionDxNet achieves impressive results, with 99.90% accuracy for colon cancer, 99.17% for lung cancer, and 99.38% for multi-class classification involving five categories. These results significantly outperform several existing state-of-the-art techniques. The findings suggest that FusionDxNet offers a highly reliable and efficient tool for aiding pathologists in early and precise cancer detection. Its implementation could streamline diagnostic workflows and contribute to better clinical outcomes through faster and more accurate decision-making.

1.1 INTRODUCTION

Cancer is one of the most daunting health challenges worldwide, standing as a significant factor in mortality among males and females across diverse populations. Among the myriad types of cancer, lung and colon cancers hold particularly grave significance due to their high incidence rates and substantial contribution to cancer-related deaths globally. Lung cancer consistently ranks as one of the most commonly diagnosed malignancies in both sexes, while colon cancer is recognized for its pervasive prevalence and considerable mortality. The aggressive biological behavior of these cancers, compounded by often late-stage detection, results in a disproportionately large number of fatalities each year, drawing attention to the immediate demand for advanced diagnostic and treatment strategies [13] [4].

Lung cancer originates from the uncontrolled proliferation of abnormal cells within lung tissues, frequently culminating in the formation of malignant tumors that

compromise respiratory function. The etiology of lung cancer is multifactorial, with smoking being the predominant risk factor, alongside advanced age and prolonged exposure to environmental carcinogens such as asbestos, radon, and industrial chemicals. Clinically, lung cancer manifests in various histological subtypes, most notably squamous cell carcinoma and adenocarcinoma, which commonly present at advanced stages when curative interventions become challenging. Similarly, colon cancer generally evolves from benign adenomatous polyps that, over time and under the influence of genetic and environmental factors, may undergo malignant transformation. These adenocarcinomas primarily arise from the epithelial lining of the colon and rectum, influenced by dietary habits, lifestyle, hereditary predisposition, and inflammatory bowel diseases [7].

Early-stage lung cancer symptoms are often subtle or nonspecific, including persistent coughing, chest discomfort, production of thick or blood-tinged sputum, and unexplained weight loss. These symptoms often look like those of benign respiratory conditions, which might make it harder to figure out what’s wrong. Timely diagnosis is critical, as treatment efficacy and patient survival rates markedly improve when lung cancer is detected at an early stage. Notably, small cell lung cancer (SCLC) constitutes approximately 15% of lung cancer cases and is characterized by rapid progression and early metastasis, whereas non-small cell lung cancer (NSCLC), comprising about 85%, exhibits relatively slower growth but remains the predominant clinical challenge [9].

Given the complexities and limitations associated with manual histopathological evaluation—including subjectivity, inter-observer variability, and resource intensiveness—there is an accelerating demand for automated, accurate, and scalable diagnostic tools. Advances in artificial intelligence, particularly convolutional neural networks (CNNs), have revolutionized the landscape of medical image analysis. CNNs possess remarkable capability in extracting hierarchical features from histopathological images, enabling precise differentiation between benign and malignant tissue architectures. This technological paradigm shift facilitates enhanced diagnostic accuracy, reduced turnaround times, and potential cost-effectiveness in clinical workflows.

Building on these transformative developments, the current research proposes a robust CNN-based framework that integrates sophisticated image preprocessing techniques alongside an optimized network architecture tailored for the classification of lung and colon cancer histopathology images. By leveraging state-of-the-art deep learning methodologies, the study aims to overcome the challenges posed by tumor heterogeneity, staining variability, and limited annotated data, thereby contributing to improved diagnostic support and ultimately better patient outcomes. **Main contributions.** In this work, our key contributions can be summarized as follows:

- We introduce **FusionDxNet**, a new framework that combines ResNet50 and MobileNet to extract complementary features from lung and colon histopathological images.
- We design a multi-stage classification pipeline that integrates Logistic Regression, SVM, a custom 1D CNN, and a hard voting ensemble to deliver robust predictions.

- The proposed approach achieves state-of-the-art performance across binary, three-class, and five-class classification tasks on the LC25000 dataset.
- We provide a detailed evaluation using accuracy, precision, recall, and F1-score, and compare our model against recent studies to demonstrate its effectiveness.
- We also discuss the limitations of the method and outline possible future directions for clinical translation.

1.2 RELATED WORK

The application of deep learning, particularly convolutional neural networks (CNNs), has revolutionized the automated classification of histopathological images for cancer diagnosis. CNNs excel at identifying subtle differences between benign and malignant tissue structures, enabling more accurate and efficient detection. This capability has been especially impactful in the diagnosis of lung and colon cancers, which remain among the leading causes of cancer-related mortality worldwide.

Table 1.1 highlights various studies showcasing the strong performance of CNN-based approaches in medical or diagnostic image analysis, focusing on the detection of lung and colorectal cancers. These works underscore the importance of optimizing network architectures, applying effective data preprocessing techniques, and developing lightweight models to enhance both diagnostic precision and practical implementation in clinical settings. In our study, we leveraged transfer learning to rapidly extract critical features from lung and colon histopathological images. This strategy enabled efficient feature representation and accurate classification, facilitating improved detection of cancerous tissues. In recent years, hybrid and transformer-based methods have been introduced for histopathological image analysis. Transformer models are effective in capturing long-range dependencies, while attention mechanisms such as MRANet[2] improve feature refinement by highlighting the most relevant regions. Lightweight CNN architectures [10, 5] have also been proposed for real-time use, balancing diagnostic accuracy with computational efficiency. Despite these advances, very few studies combine **dual CNN feature extractors with multi-classifier fusion**. Our proposed FusionDxNet addresses this gap by leveraging complementary strengths from multiple models, leading to improved generalization and higher classification accuracy.

1.3 MATERIALS AND METHODS

The LC25000 histopathology image collection and pre-trained networks like as ResNet50 and MobileNet are among the study's resources. To fuse features for categorization, the system uses a dual-stream feature extraction procedure.

1.3.1 Dataset

The LC25000 dataset [3] has become a widely adopted benchmark in lung and colon cancer classification research. Comprising 25,000 high-resolution histopathological

Table 1.1: Literature Review Summary

papers	Methodology	Contribution	Limitation
Mahmud et al. (2025)[6]	a hybrid deep learning model that integrates a modified InceptionV3 and a lightweight CNN to classify lung cancer using both CT and histopathological images, with built-in interpretability	A new hybrid, multi-modality model for lung cancer diagnosis and the integration of explainable AI, resulting in high performance	use of a specific dataset, potential computational demands, and the absence of clinical validation
Shahadat et al. (2024)[10]	Lightweight 1D CNN with Squeeze-and-Excitation blocks	Introduced a mobile-compatible, efficient model suitable for real-time deployment	Extremely high accuracy may suggest overfitting; it lacks cross-dataset testing
Bala et al. (2024)[2]	Modified Residual Attention Network (MRANet) for 2-class, 3-class, and 5-class classification	Attention-based feature refinement improved performance across multiple cancer types	Computationally intensive; not optimized for low-resource devices
Wadekar & Singh (2023)[13]	Modified VGG19 with data augmentation for lung cancer classification using histopathological images	Demonstrated effective use of transfer learning with image preprocessing	Focused only on lung cancer; lacks lightweight deployment analysis
Hadiyoso et al. (2022)[4]	CNN with CLAHE preprocessing on LC25000 dataset	Highlighted the value of image enhancement techniques for improved classification	Used only VGG16; limited discussion on model generalization
Mangal et al. (2020)[7]	Shallow CNN trained on LC25000 for classifying benign and malignant tissues in the lung and colon	Provided a simple yet effective architecture for multi-class classification	No use of attention mechanisms or feature fusion

images, each measuring 768×768 pixels, the dataset offers a diverse representation of cancerous and non-cancerous tissues. It is systematically divided into five balanced categories: lung adenocarcinoma, lung squamous cell carcinoma, benign lung tissue, colon adenocarcinoma, and benign colon tissue. This uniform class distribution makes the dataset particularly valuable for training and evaluating deep learning models in multi-class classification tasks involving both lung and colon cancer.

Figure 1.1 showcases representative samples from the dataset, enhanced through various augmentation techniques. These augmentations were applied to a condensed version of the original dataset to ensure clinical significance while introducing suf-

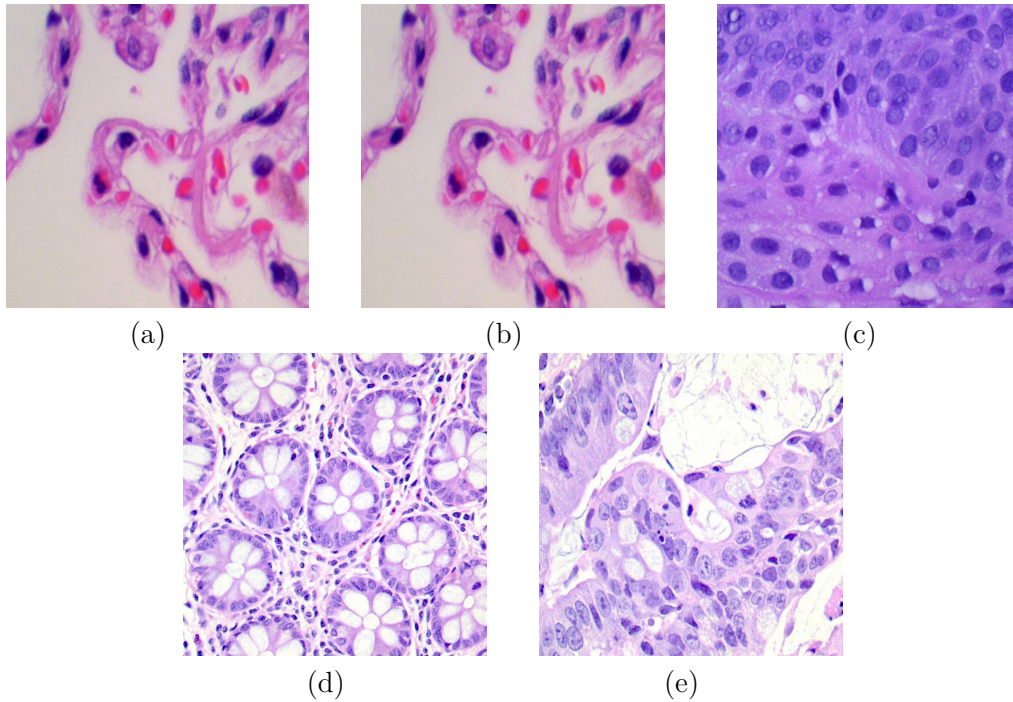


Figure 1.1: Histopathological image (a) lung_aca (b) lung_n (c) lung_scc (d) colon_n (e) colon_aca

ficient variability for robust learning. The LC25000 dataset has gained widespread adoption in medical image classification tasks, particularly for training and evaluating convolutional neural networks (CNNs), due to its class balance, accessibility, and consistent performance in cancer detection research.

In this study, 80% of the dataset was randomly selected for training purposes, while the remaining 20% was reserved for testing and final evaluation. To monitor performance during training, 10% of the training data was further separated and used as a validation set.

1.3.2 Method Overview

The overall workflow of FusionDxNet is presented in Fig. 1.2. It operates in three main stages:

- **Stage 1 – Feature Extraction:** Images are processed through pretrained CNN backbones (ResNet50 and MobileNet), each capturing distinct feature representations.
- **Stage 2 – Feature Fusion:** The extracted features are concatenated into a unified vector that combines both deep hierarchical patterns and lightweight descriptors.
- **Stage 3 – Classification:** The fused and individual feature sets are classified

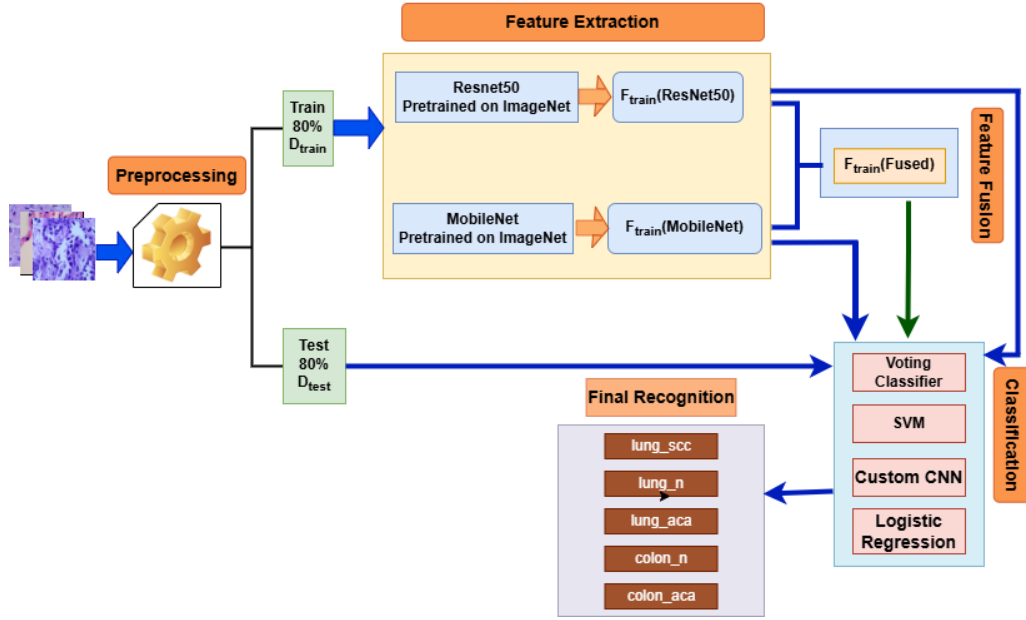


Figure 1.2: Proposed Network Architecture

using Logistic Regression, SVM, a custom 1D CNN, and a hard voting ensemble to produce final predictions.

1.3.3 Proposed Methodology

Our proposed architecture, FusionDxNet, is designed with a dual-module structure: a feature extraction unit and a classification engine. At its core, FusionDxNet leverages the power of multiple pretrained deep learning models to extract meaningful and high-dimensional features from histopathological images. These models act as intelligent visual analyzers, capable of identifying complex patterns and textures that are often difficult for the human eye or traditional techniques to distinguish.

The extracted features from these individual models are not directly used for final classification in all cases. Instead, FusionDxNet provides two complementary pathways to evaluate model performance. First, each model independently passes its extracted features to a dedicated classification layer, forming the single-model evaluation stream, shown as the blue path in Fig. 1.2. This allows us to assess how each standalone model performs on its own in detecting and classifying the cancer types from the test data.

The second and more robust approach is the fusion-based evaluation, represented by the green path in Fig. 1.2. Here, the outputs (i.e., the learned features) from multiple pretrained models are intelligently combined using a feature fusion strategy. This creates a unified, enriched feature representation that captures a broader set of visual cues. The fused features are then fed into a final classification layer, providing an ensemble decision that benefits from the strengths of all contributing models.

Table 1.2: Performance comparison of different feature extractors and classifiers

Classes	Feature Extractor	Classifier	Accuracy	Macro			Weighted		
				Precision	Recall	F1 Score	Precision	Recall	F1 Score
2	ResNet50	Custom Classifier	99.85	99.85	99.85	99.85	99.85	99.85	99.85
	ResNet50	MLP	99.75	99.75	99.75	99.75	99.75	99.75	99.75
	MobileNet	Custom Classifier	99.70	99.70	99.70	99.70	99.70	99.70	99.70
	MobileNet	Logistic Regression	99.90	99.90	99.90	99.90	99.90	99.90	99.90
3	ResNet50 & MobileNet	Logistic Regression	99.90	99.90	99.90	99.90	99.90	99.90	99.90
	MobileNet	Voting Classifier	98.63	98.63	98.63	98.63	98.63	98.63	98.63
	MobileNet	Custom Classifier	98.87	98.87	98.87	98.87	98.87	98.87	98.87
	ResNet50	SVM	99.07	99.07	99.07	99.07	99.07	99.07	99.07
5	ResNet50	Logistic Regression	98.93	98.93	98.93	98.93	98.93	98.93	98.93
	ResNet50 & MobileNet	SVM	99.17	99.17	99.17	99.17	99.17	99.17	99.17
	MobileNet	Voting Classifier	99.12	99.12	99.12	99.12	99.12	99.12	99.12
	MobileNet	MLP	99.16	99.16	99.16	99.16	99.16	99.16	99.16
5	ResNet50	SVM	99.26	99.26	99.26	99.26	99.26	99.26	99.26
	ResNet50	MLP	99.38	99.38	99.38	99.38	99.38	99.38	99.38
	ResNet50 & MobileNet	Voting Classifier	99.38	99.38	99.38	99.38	99.38	99.38	99.38
	ResNet50 & MobileNet	Voting Classifier	99.38	99.38	99.38	99.38	99.38	99.38	99.38

This fusion process enhances robustness, improves generalization, and often results in better diagnostic accuracy than any single model alone.

For training and evaluation, the dataset is systematically divided into three subsets: training, validation, and test sets. The training set is used to optimize the classifier, the validation set helps tune hyperparameters and prevent overfitting, and the final evaluation is conducted on the test set to measure real-world performance. The entire end-to-end workflow, from feature extraction to classification and evaluation, is clearly illustrated in Fig. 1.2, highlighting both the single-model and fusion-based paths.

1.3.4 Feature Extractor

In this research, we harness the power of two cutting-edge convolutional neural network (CNN) architectures—ResNet50 and MobileNet—as the core engines for feature extraction. Both models are employed in their pretrained form, having been originally trained on the ImageNet dataset, which comprises over a million images across a thousand categories. This prior training equips them with a rich understanding of fundamental visual patterns such as edges, textures, and object structures.

Instead of training these deep networks from the ground up, we adopt a transfer learning approach. By retaining the pretrained weights and filters learned during ImageNet training, the models can quickly adapt to our specialized domain of lung and colon histopathological image analysis. This strategy not only accelerates convergence but also improves accuracy, especially when working with limited labeled medical data.

Each model—ResNet50 and MobileNet—extracts a distinct set of high-level features from the input images, reflecting their unique architectural strengths. While ResNet50 excels in capturing deep hierarchical patterns through residual learning, MobileNet provides lightweight, efficient feature extraction optimized for speed and low computational cost.

To maximize the diagnostic potential, we employ a feature fusion strategy that integrates the outputs from both models into a unified feature representation. This fused vector captures a more holistic view of the histopathological structures, enhancing the discriminative power of the classification system. The entire workflow, including loading pretrained models, feature extraction, and fusion, is systematically outlined in Algorithm 1.

Algorithm 1 Dual-Stream Feature Extraction Process

Input: Image dataset D , Pretrained networks: ResNet50, MobileNet

Output: Feature representations: F_{ResNet50} , $F_{\text{MobileNet}}$, F_{Fused}

- 1: Divide D into training and testing subsets: $D_{\text{train}}, D_{\text{test}} \leftarrow \text{Split80to20}(D)$
 - 2: Apply ResNet50 on training data: $F_{\text{train(ResNet50)}} \leftarrow \text{ResNet50}(D_{\text{train}})$
 - 3: Apply MobileNet on training data: $F_{\text{train(MobileNet)}} \leftarrow \text{MobileNet}(D_{\text{train}})$
 - 4: Merge feature vectors: $F_{\text{train(Fused)}} \leftarrow \text{Concatenate}(F_{\text{train(ResNet50)}}, F_{\text{train(MobileNet)}})$
 - 5: **return** $F_{\text{train(ResNet50)}}, F_{\text{train(MobileNet)}}, F_{\text{train(Fused)}}$
-

1.3.5 Classification

We generate three distinct sets of features — $\mathbf{F}_{\text{train}}(\text{ResNet50})$, $\mathbf{F}_{\text{train}}(\text{MobileNet})$, and $\mathbf{F}_{\text{train}}(\text{Fused})$ — which serve as inputs for training our classification models. To comprehensively assess the quality and discriminative power of these feature representations, we utilize three separate classification algorithms, each bringing its own strengths in learning patterns and defining decision boundaries. The complete classification workflow is outlined in Algorithm 2.

Algorithm 2 Training Multiple Classifiers

Input: Feature sets $\mathcal{F}$, Classifiers $\mathcal{C} = \{\text{Logistic Regression (LG)}, \text{Custom CNN (CCNN)}, \text{textSupportVectorMachine(SVM)}, \text{Voting Classifier (V)}\}$, Corresponding labels L

Output: Trained models $\{LG^*, SVM^*, CCNN^*, V^*\}$

```

1: for all feature set  $f \in \mathcal{F}$  do
2:   for all classifier  $c \in \mathcal{C}$  do
3:     Train classifier  $c$  on feature set  $f$  with labels  $L$ :  $c.\text{train}(f, L)$ 
4:   end for
5: end for
6: return all trained classifiers:  $LG^*, SVM^*, CCNN^*, V^*$ 

```

1.3.5.1 Support Vector Machine:

SVM is a margin-based learning algorithm designed to find the optimal hyperplane that maximally separates classes by the widest margin. It relies on support vectors—critical training points that define this boundary. Through the use of kernel functions, SVMs can efficiently handle both linear and nonlinear classification tasks by projecting data into higher-dimensional feature spaces.

1.3.5.2 Custom CNN Model

A bespoke convolutional neural network built with three sequential Conv1D layers featuring increasing filter counts (32, 64, 128). Each convolutional layer is paired with a MaxPooling1D operation to downsample feature maps. The extracted features are then flattened and fed into a fully connected dense layer with 256 neurons, incorporating dropout (rate 0.5) for regularization. The model concludes with a softmax activation layer that outputs class probabilities.

1.3.5.3 Voting Classifier:

An ensemble model that integrates predictions from multiple classifiers, including logistic regression, linear support vector machine (LSVM), and perceptron. Additionally, it combines predictions from a broader set of classifiers, such as logistic regression, LSVM, perceptron, SVM with quadratic penalties, and SVM with a smooth quadratic loss, utilizing a hard voting mechanism.

1.4 EXPERIMENTATION AND PERFORMANCE ANALYSIS

In this section to show the suggested FusionDxNet model is evaluated in detail. For two-class situations, the study demonstrates the model's exceptional performance, with an accuracy of up to 99.90%. This section also highlights the model's efficacy in lung and colon cancer classification by contrasting its output with that of various other cutting-edge techniques.

1.4.1 Model's Performance Evaluation

Using a number of important criteria, each of which provided insightful performance information, we assessed our model's efficacy in ASD screening from face photos.

1.4.1.1 Accuracy

It measures the proportion of total correct predictions made by the model, serving as a straightforward indicator of overall prediction correctness.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (1.1)$$

1.4.1.2 F1 Score

The F1 score provides a harmonic mean between precision and recall, balancing the trade-off between false positives and false negatives to give a single metric for model reliability.

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1.2)$$

1.4.1.3 Precision

Precision reflects the accuracy of positive predictions by calculating the fraction of true positive results among all instances predicted as positive.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (1.3)$$

1.4.1.4 Recall

Recall, also referred to as sensitivity or the true positive rate, indicates the model's ability to correctly identify all actual positive cases, minimizing missed detections.

Recall (Sensitivity or True Positive Rate):

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (1.4)$$

1.4.2 Performance Evaluation

In this study, we employed two convolutional neural network architectures, namely ResNet50 and MobileNet, both pretrained on the ImageNet dataset, to perform feature extraction. These individual feature sets were further combined through a fusion process to create an additional comprehensive feature representation. Subsequently, four classifiers—logistic regression, a custom 1D CNN, support vector machine (SVM), and a voting ensemble—were applied to classify the extracted features. Across various experimental trials, the voting classifier demonstrated superior performance in accuracy and other key evaluation metrics, despite having a lower number of parameters compared to the other models. A detailed comparison of these results is provided in Table 1.2, while the confusion matrix corresponding to our best-performing model is presented in Fig. 1.3.

Table 1.3: Comparison with Existing Literature

Authors	Methods	Accuracy (%)	No. of Classes
S. Hadiyoso <i>et al.</i> [4]	VGG16 with CLAHE	98.96	5
M. A. Hasan <i>et al.</i> [5]	Lightweight CNN	99.20	5
R. R. Swain <i>et al.</i> [12]	ML algorithms	95.00	2
M. Al-Mamun Provath <i>et al.</i> [1]	CNN	97.00	5
M. Masud <i>et al.</i> [8]	Deep Learning Model	96.33	5
A. Sultana <i>et al.</i> [11]	Hybrid CNN-SVM	99.13	5
S. Mangal <i>et al.</i> [7]	CNN	96.00	3
S. Wadekar <i>et al.</i> [13]	VGG16	97.73	3
D. Bala <i>et al.</i> [2]	MRANet	99.30, 96.63, 97.56	2, 3, 5
M.R. Mahmud <i>et al.</i> [6]	MRANet	99.30, 96.63, 97.56	2, 3, 5
Proposed	FusionDxNet	99.90, 99.17, 99.38	2, 3, 5

Among all the experimental configurations explored, the combination of features extracted from MobileNet and classified using a voting ensemble delivered the most promising results, achieving top-tier performance with minimal model complexity. Specifically, this approach attained an impressive 99.90% accuracy for colon cancer detection, 99.17% accuracy for lung cancer classification, and 99.38% accuracy when handling both cancer types simultaneously. To ensure the robustness of our approach, we evaluated it using the same dataset and benchmarked the performance against existing methods (Table 1.3). The results indicate that our proposed model significantly outperforms previous techniques in terms of classification accuracy.

1.5 LIMITATION & FUTURE WORK

Although FusionDxNet shows excellent classification performance, several challenges remain.

First, the dataset images have medium resolution (768×768 pixels), which may limit the detection of very fine tissue structures.

Second, since experiments were conducted only on the LC25000 dataset, the generalizability of the model to data from different hospitals or imaging devices has not yet been validated. Third, the classifiers used default hyperparameters, and the CNN backbones were not fine-tuned for medical images, which could restrict adaptability to clinical practice.

Third, the classifiers implemented in our pipeline—logistic regression, support vector machine (SVM), a custom CNN, and a voting-based ensemble—were trained using default hyperparameters. Without thorough optimization, these models may not be performing at their maximum potential.

Additionally, the pre-trained CNN models, ResNet50 and MobileNet, were used solely as fixed feature extractors. Since they were not fine-tuned on our domain-specific data, they may not have adapted fully to the unique characteristics of medical imaging, potentially limiting their effectiveness.

Future improvements may be achieved by integrating higher-resolution and more diverse image datasets, which could provide richer visual patterns for the models to learn from. Moreover, implementing systematic hyperparameter tuning techniques—such as grid search, random search, or Bayesian optimization—can help maximize the performance of the classifiers. Lastly, fine-tuning the pre-trained CNNs on the target domain could allow the models to learn more relevant features specific to cancer detection tasks.

Together, these strategies could lead to more accurate, robust, and clinically applicable diagnostic tools.

1.6 CONCLUSION AND FUTURE WORK

In this study, we introduced FusionDxNet, a transfer learning-based framework designed for the classification of lung and colon cancer using histopathological images. The approach leveraged ResNet50 and MobileNet as pretrained feature extractors and evaluated them with different classifiers, including Logistic Regression, a custom 1D CNN, and SVM. Experimental outcomes demonstrated the robustness of the framework: 99.90% accuracy in binary classification with MobileNet–Logistic Regression, 99.17% accuracy in three-class classification with the ResNet50–MobileNet–SVM setup, and 99.38% accuracy in the five-class task using the ResNet50–SVM pipeline.

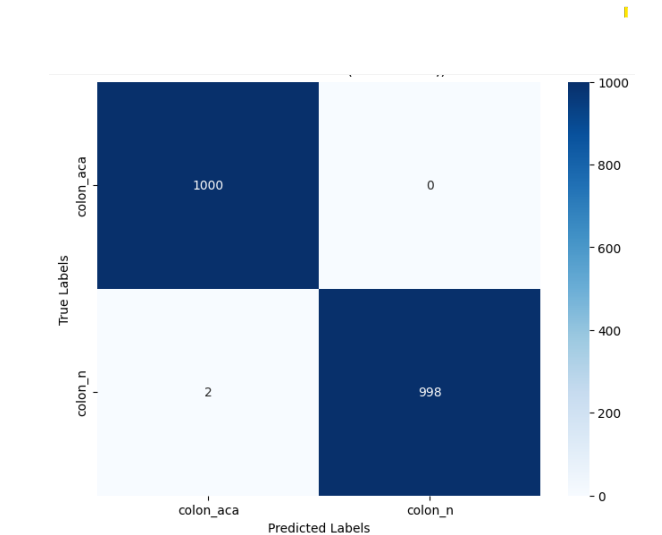
These results highlight the potential of combining multiple CNN-based feature representations with diverse classification strategies to achieve high diagnostic reliability. The inclusion of a 1D CNN classifier further enriched feature interpretation, allowing the system to capture complex non-linear patterns within medical images.

For future advancement, the model can be improved through fine-tuning the pre-trained CNNs, adopting systematic hyperparameter tuning methods, and employing higher-resolution, multi-center datasets. Additional research could also extend FusionDxNet to multimodal imaging data and assess its integration into real-world clinical workflows, which would further enhance accuracy, scalability, and clinical applicability.

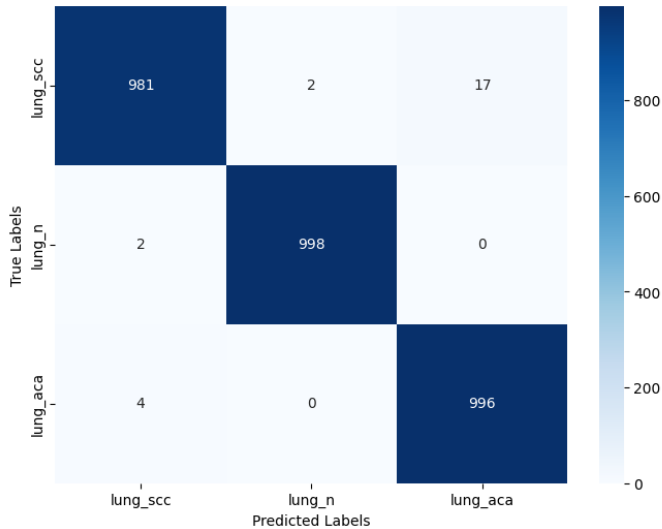
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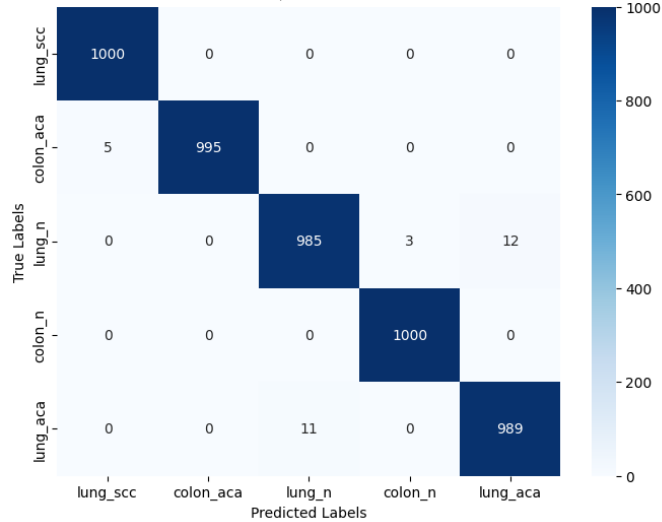
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(a) Feature extractor: MobileNet
Classifier: Logistic Regression



(b) Feature extractor: MobileNet, Res-
Net50 , Classifier: SVM



(c) Feature extractor: Resnet50
Classifier: SVM

Figure 1.3: Confusion matrix of the best results